

Roll No.

TCE-601

B. TECH. (CE) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

ENVIRONMENTAL ENGINEERING-II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Discuss the relative merits of the separate and the combined systems of sewage and give the favorable condition for the adoption of each one of them. 10 Marks (CO1)

OR

- (b) Briefly discuss the method one will adopt for working out the design capacities of various laterals, branch sewers, and main sewers of a sewerage system. 10 Marks (CO1)
2. (a) Differentiate between 'Sewage' and 'Drainage'. Discuss the Rational formulae for calculating the peak drainage from a given catchment and reaching a particular storm water drain upto a particular point.

10 Marks (CO1)

P. T. O.

OR

- (b) A certain district of a city has a projected population of 50,000 residing over an area of 40 hectares. Find the design discharge for the sewer line, for the following data : 10 Marks (CO1)

- (i) Rate of water supply = 200 l/d/person
- (ii) Average impermeability coefficient for the entire area = 0.3
- (iii) Time of concentration = 50 min

The sewer line is to be designed for a flow equivalent to the wet weather flow (W.W.F.) plus twice the dry weather flow (D. W. F.). Assume that 75% of water supply reaches in sewer as wastewater.

3. (a) Briefly explain the method of finding out the sizes of sanitary sewers when the discharge to be carried through them are known.

10 Marks (CO2)

OR

- (b) A rectangular sewer with width 1.5 times its depth is hydraulically equivalent to a circular one. Illustrate the relation between the width of the rectangular sewer and the diameter of the circular sewer is $D = 1.043 B$, where D is the diameter of circular pipe and B is the width of rectangular section. 10 Marks (CO2)

4. (a) Discuss the various factors that affect the hydraulics of sewer lines.

10 Marks (CO2)

OR

- (b) Explain the importance of the following in the design of sewers :

10 Marks (CO2)

- (i) Self-cleansing velocity
- (ii) Non-scouring velocity

(3)

5. (a) Describe the various components of sewerage systems in order of construction. 10 Marks (CO2)

OR

- (b) List out various types of sewers based on their cross-sections with highlighting the place of applications of each type. 10 Marks (CO2)

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TCE-602

**B. TECH. (CE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April/May, 2022**

REINFORCED CEMENT CONCRETE-II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Present a comparative discussion between reinforced concrete vs. prestressed concrete members in any *five* key aspects. 10 Marks (CO1)

OR

- (b) Determine the stresses at top and bottom fibers of the given rectangular PSC beam at quarter span with a cross section of 200×500 mm and 6 m long span and is provided with 3-5 mm dia steel wires positioned at 25 mm from top fiber of the beam and 15-5 mm dia steel wires whose resultant force is concentrated at 65 mm from the soffit of the beam. A net prestress of 840 N/mm^2 is applied in-all the wires after considering all the losses. 10 Marks (CO1)

P. T. O.

2. (a) Explain the various post-tensioning methods of practice in prestressed concrete member production, with an emphasis on the pros and cons of each method. 10 Marks (CO1)

OR

- (b) A PSC beam (span 10 m) is prestressed by a parabolic cable having an eccentricity of 100 mm at the center of the span and reducing to zero at the supports is used. The effective force in the cable being 180 kN. The beam supports a total uniformly distributed load of 5 kN/m which includes the self-weight of the member. Estimate the % reduction in the principal tension in comparison with the case of no prestress.

10 Marks (CO1)

3. (a) A rectangular concrete beam, 300 mm deep and 200 mm wide is prestressed by 15 Nos of 5 mm diameter wires located 65 mm from the bottom of the beam and three 5 mm wires located 25 mm from the top of the beam. If the wires are initially tensioned to a stress of 840 N/mm^2 , calculate the percentage loss of stress in steel immediately after the transfer allowing for the loss of stress due to elastic deformation of concrete only. $E_s = 210 \text{ kN/mm}^2$, $E_c = 31.5 \text{ kN/mm}^2$.

10 Marks (CO1)

OR

- (b) A concrete beam of 10 m span, 100 mm wide and 300 mm deep, is prestressed by 3 cables. The area of each cable is 200 mm^2 and the

initial stress in the cable is 1200 N/mm^2 . Cable 1 is parabolic with an eccentricity of 50 mm above the centroid at the supports and 50 mm below at the center of the span.

Cable 2 is also parabolic with zero eccentricity at the supports and 50 mm below the centroid at the center of span.

Cable 3 is straight with uniform eccentricity of 50 mm below the centroid.

If the cables are tensioned from one end only, estimate the percentage loss of stress in each cable due to friction. Assume $\mu = 0.35$ and $k = 0.005/\text{m}$.

10 Marks (CO1)

4. (a) Discuss the various assumptions involved in the structural analysis of continuous beams by elastic theory, and their implications on the design of RC continuous beams.

10 Marks (CO2)

OR

- (b) Develop the design bending moment envelop of a propped cantilever beam of span 8 m and simply supported at the right end and carrying ultimate uniformly distributed load of 20 kN/m with full redistribution of 30 per cent as per IS 456.

10 Marks (CO2)

P. T. O.

5. (a) Establish the “shifting length” of the of “point of contraflexure” in developing the bending moment envelope for a RC fixed beam clamped at both ends and carrying ultimate uniformly distributed load of 24 kN/m with a redistribution of 25 per cent as per IS 456 guidelines.

10 Marks (CO2)

OR

- (b) Derive expressions for variation of shear force, bending moments and twisting moment diagrams along the span of a semi-circular beam fixed at the ends supporting a uniformly loaded semi-circular floor slab.

10 Marks (CO2)

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TCE-603

B. TECH. (CIVIL ENGINEERING) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

WATER RESOURCES ENGINEERING—II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) State the factors that are to be considered while selecting suitable site for a given dam. 10 Marks (CO1)

OR

- (b) An elementary profile of a concrete gravity dam of triangular section has maximum water level up to the top of the dam which is 50 m. high d/s slope has a better of 0.8 H : 1.0. There is no tail water.

10 Marks (CO1)

Analyze the dam section and determine

Factor of safety against overturning

Factor of safety against sliding

Uplift pr. intensity coefficient, acting of 100% area = 0.5

Coeff. of friction between base and foundation = 0.67

P. T. O.

2. (a) Explain briefly with neat sketch the different forces that may act on a gravity dam. 10 Marks (CO1)

OR

- (b) A gravity dam has the following dimension : 10 Marks (CO1)

Height of the dam = 100 m

Freeboard = 1 m

u/s slope = 0.15 H : 1 V

taking K_h = 0.1

Determine hydrostatic earthquake pressure and its moment at the base of dam.

3. (a) Describe briefly the various design considerations for designing the earthen dam. 10 Marks (CO2)

OR

- (b) Determine the phreatic line for an earthen dam homogeneous in section from the following data : 10 Marks (CO2)

Height of dam = 20 m

Free board = 2.0 m

Coefficient of permeability of material of the dam = 4×10^{-4} cm/sec

Top width = 4 m

Upstream slope = 3 : 1

Downstream slope = 3 : 1

A horizontal filter provided extends to 30 m. from the toe point of the dam.

And also find discharge per metre length of the earthen dam.

(3)

4. (a) Draw a neat sketch (section) of an earthen dam and name the various parts. 10 Marks (CO2)

OR

- (b) Determine the factor of safety of downstream slope of an earthen dam (homogeneous section). 10 Marks (CO2)

Drawn to a scale of 1 cm = 4 m

Length of slip circle arc = 15 cm

Total area of N-rectangle = 16.5 cm^2

Total area of T-rectangle = 7.0 cm^2

Total area of U-rectangle = 5 cm^2

Angle $\phi = 26^\circ$

Cohesion $c = 20 \text{ kN/m}^2$

Specific weight of soil 18 kN/m^3

5. (a) Explain the thin cylinder theory of design of an arch dam. What are the limitations of this theory? 10 Marks (CO1)

OR

- (b) A 10 m high concrete dam having trapezoidal section has top width 1 m and bottom width 8.25 m. The face of the dam exposed to water has a slope 0.1 H : 1 V. On the reservoir side water stand up to the top. Assuming unit weight of conc. is 24 kN/m^3 . 10 Marks (CO2)

Coeff. of friction is 0.75 and allowable shear stress = 4000 kN/m^2 , and taking in to account only the weight of the dam, water pressure and uplift pressure, calculate the factor of safety against overturning, against sliding and shear friction factor.

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TCE-604

B. TECH. (CIVIL ENGINEERING) (SIXTH SEMESTER) MID SEMESTER EXAMINATION, April/May, 2022

QUANTITY ESTIMATION AND COSTING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Define estimate and describe different types of estimate in civil engineering works. 10 Marks (CO1)

OR

- (b) A building project with total plinth area of all building is 800 sqm. From following data : 10 Marks (CO1)

- (i) Plinth area rate Rupees 4,500 per sq. m.
- (ii) Cost of water supply @ $7\frac{1}{2}$ % of cost of building.
- (iii) Cost of Sanitary and Electrical installations each @ $7\frac{1}{2}$ % of cost of building.
- (iv) Cost of architectural features @ 1% of building cost.

P. T. O.

(v) Cost of roads and lawns @ 5% of building cost.

(vi) Cost of P. S. and contingencies @ 4% of building cost.

Calculate the total cost of building project.

2. (a) Describe the two systems adopted in taking quantities in preparing estimate of civil engineering works. 10 Marks (CO1)

OR

- (b) The plinth area of an apartment is 500 sq. m. Calculate the total cost of building from the following data : 10 Marks (CO1)

(i) Rate of construction = ₹ 1,230/-per m³.

(ii) The height of apartment = 16.25 m.

(iii) Water Supply, Sanitary and Electrical installations each at 6% of building cost.

(iv) Architectural appearance @ 1% of building cost.

(v) Unforeseen item @ 2% of building cost.

P. S. and contingencies @ 4% of building.

3. (a) Explain unit of measurement and use of different units of measurement in civil engineering works. 10 Marks (CO1)

OR

- (b) The plinth area and plinth area rate of a residential building are 100 sq. m. and Rupees. 5,000/ respectively. Predict the total cost of building assuming suitable provisions. 10 Marks (CO1)

(3)

4. (a) Explain overhead cost and classify under different heads.

10 Marks (CO2)

OR

- (b) Assuming current schedule of rates for materials and labour. Calculate data for the Plain cement concrete 1 : 5 : 10 in foundation.

10 Marks (CO2)

5. (a) Define rate analysis and identify factors that affecting rate analysis.

10 Marks (CO2)

OR

- (b) Assuming current schedule of rates for materials and labour. Calculate data for the Plastering walls in cm 1 : 5 – 12 mm thick.

Roll No.

TCE-605

B. TECH. (CE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April/May, 2022
TRANSPORTATION ENGINEERING-I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Discuss the merits and demerits of various modes of transportation.

10 Marks (CO1)

OR

- (b) The design speed of a highway is 80 kmph. There is a horizontal curve of radius 200 m on a certain locality. Safe limit of transverse coefficient of friction is 0.15 :

(i) Calculate the superelevation required to maintain this speed,

(ii) If the maximum superelevation is 0.07 is not be exceeded,

calculate the maximum allowable speed on this horizontal curve as it is not possible to increase the radius.

10 Marks (CO1)

2. (a) Discuss the various methods of classifying roads. Briefly, outline the classification based on location and function as suggested in the Nagpur road plan.

10 Marks (CO1)

P. T. O.

(2)

OR

(b) Calculate the values of the following :

- (i) Head light sight distance and
- (ii) Intermediate sight distance for a highway with a design speed of 65 km/hr.

Assume suitably all the data required.

10 Marks (CO1)

3. (a) Discuss the different factors affecting the alignment of highway.

10 Marks (CO1)

OR

- (b) An ascending gradient 1 in 60 meets a descending gradient of 1 in 50.
Find the length of vertical summit curve for a stopping sight distance of 180 m.

10 Marks (CO1)

4. (a) List different tests on road aggregates and mention the principle and uses of each test.

10 Marks (CO2)

OR

- (b) Describe the desirable properties of subgrade soil.

10 Marks (CO2)

5. (a) Describe the various tests carried out on bitumen. Briefly mention the principle and uses of each test.

10 Marks (CO2)

OR

- (b) Describe the desirable properties of aggregates to be used in different types of pavement construction.

10 Marks (CO2)